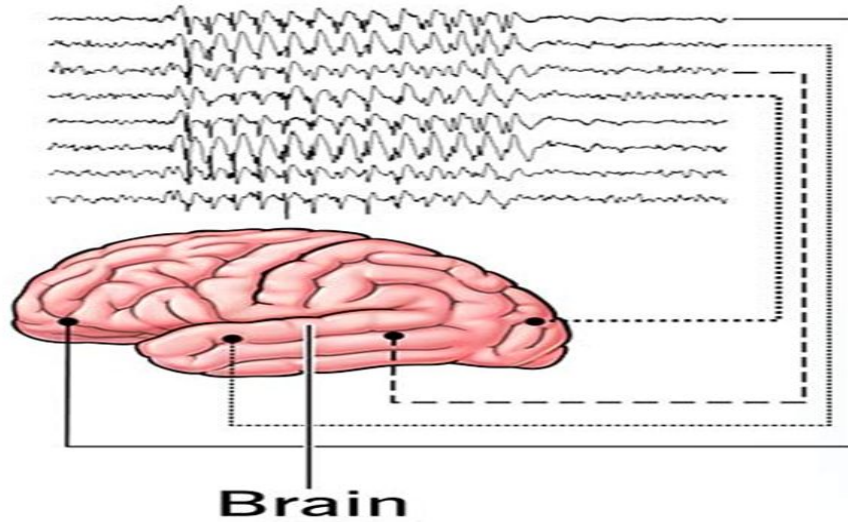


Electroencephalogram (EEG)

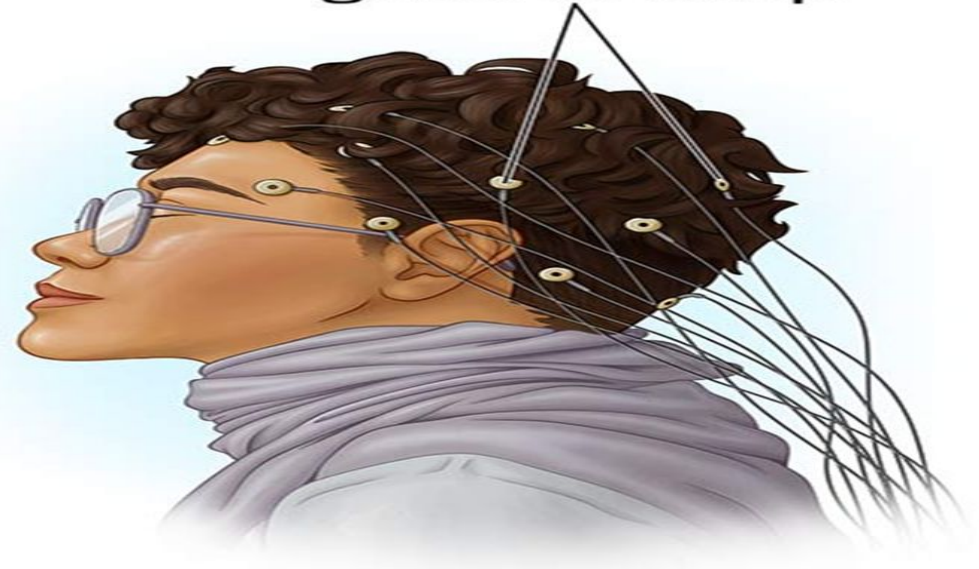
Electroencephalogram (EEG)

EEG (scan of brainwaves)




Cleveland
Clinic
©2023

Electrodes
glued to scalp



DR ABDUL SAMAD KHAN
SENIOR LECTURER-DIKIOHS
DOW UNIVERSITY OF HEALTH SCIENCES

Electroencephalogram (EEG)

The electroencephalogram (EEG) is a recording of the electrical activity of the brain from the scalp.

The first recordings were made by Hans Berger in 1929



Electroencephalography

An electroencephalogram (EEG) measures electrical activity in the brain to track how brain wave patterns shift during different stages of sleep.

EEG is used to measure brain rhythms through recording electrical activity at the scalp, making it a non-invasive and painless tool to detect brain activity.

EEG detects the general activity of the cortex by measuring small (~ 10 μV) fluctuations in voltage.

This technique does not measure electrical activity of a single neuron, but rather measures the electrical activity of a large population of neurons, on the order of thousands of cells, in order to generate a signal that is large enough to detect.

Why is an EEG done?

Healthcare providers often order this test if a person have **seizures** or if the person suspect **epilepsy**. Doctor may also recommend one if you have symptoms like:

- Confusion
- Fainting
- Memory loss
- Changes in behavior
- EEGs can also monitor brain activity in certain conditions, including:
 - Alzheimer's disease or dementia
 - Brain infections like encephalitis
 - Brain tumors
 - Coma
 - Traumatic brain injury

Types of EEG tests

Doctors order different types of EEGs depending on the medical issue. These include:

Routine EEG: This may use flashing lights or changes in breathing to record brain activity.

Prolonged EEG: This records brain activity for about an hour or longer. Providers may use it to see if symptoms like memory lapses link to seizures.

Ambulatory EEG: This lets you go about your normal day while a small recorder tracks brain activity for one or more days.

Video EEG: This records brain activity along with video and audio. It helps providers see what you're doing during a seizure. It's also called EEG monitoring, EEG telemetry or video EEG monitoring.

Sleep EEG: Sometimes done with a sleep study (**polysomnogram**), this helps gather more information during sleep. Providers may order it if they suspect a **sleep disorder** linked to epilepsy.

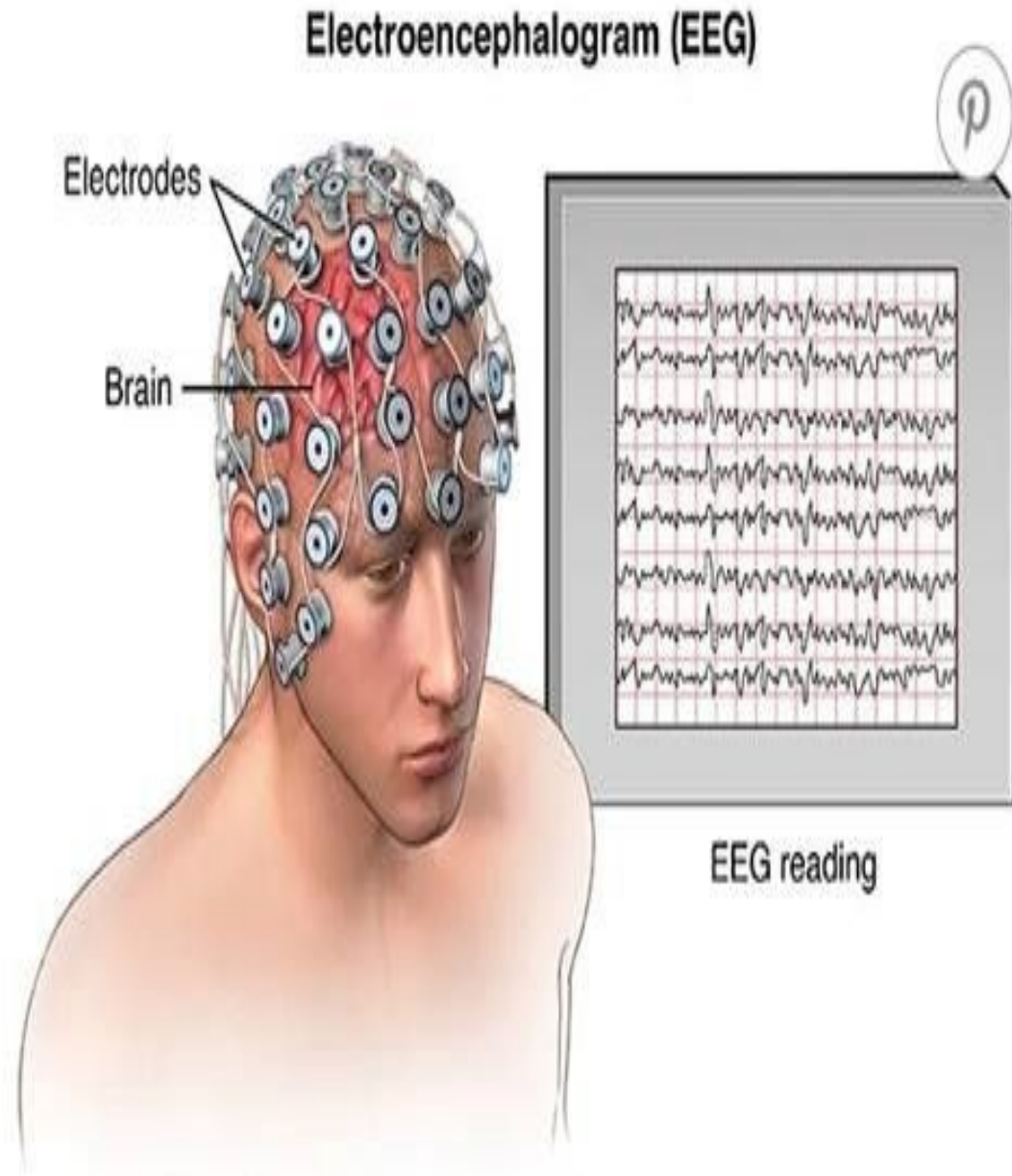
An EEG reading depicts differences in both **amplitude** and **frequency** over time.

An increase in **amplitude** on an EEG indicates a higher degree of synchronous activity of cortical neurons, whereas the **frequency** of an EEG indicates how often neural synchrony occurs.

EEG is used to measure patterns of brain activity during different behavioral states and is used in both research and in the diagnosis of various neurological conditions such as sleep disorders and **epilepsy**.

How does an EEG work?

- This test measures the electrical signals (impulses) that pass between your brain cells.
- Electrodes on the scalp pick up these signals through small wires. The process works like this:
 1. The electrodes send the signals to an EEG machine.
 2. The machine records brain waves and displays them on a screen.
 3. Brain wave patterns change when you are awake, asleep or when something affects your brain.
 4. The image on the screen is your EEG.
 5. Your healthcare provider reviews the image to look for abnormal patterns.



How long does an EEG take?

A routine EEG may take 20 to 30 minutes to complete. The time depends on the type. But your provider will tell you in advance so you know what to expect.

EEG test side effects

This test rarely causes side effects.

Some people feel dizzy while taking deep breaths.

If patient have certain types of epilepsy, deep breathing (hyperventilation) or flashing lights (photic stimulation) may trigger a seizure.

Patient may also notice minor and temporary hair loss or skin irritation where the electrodes were placed.

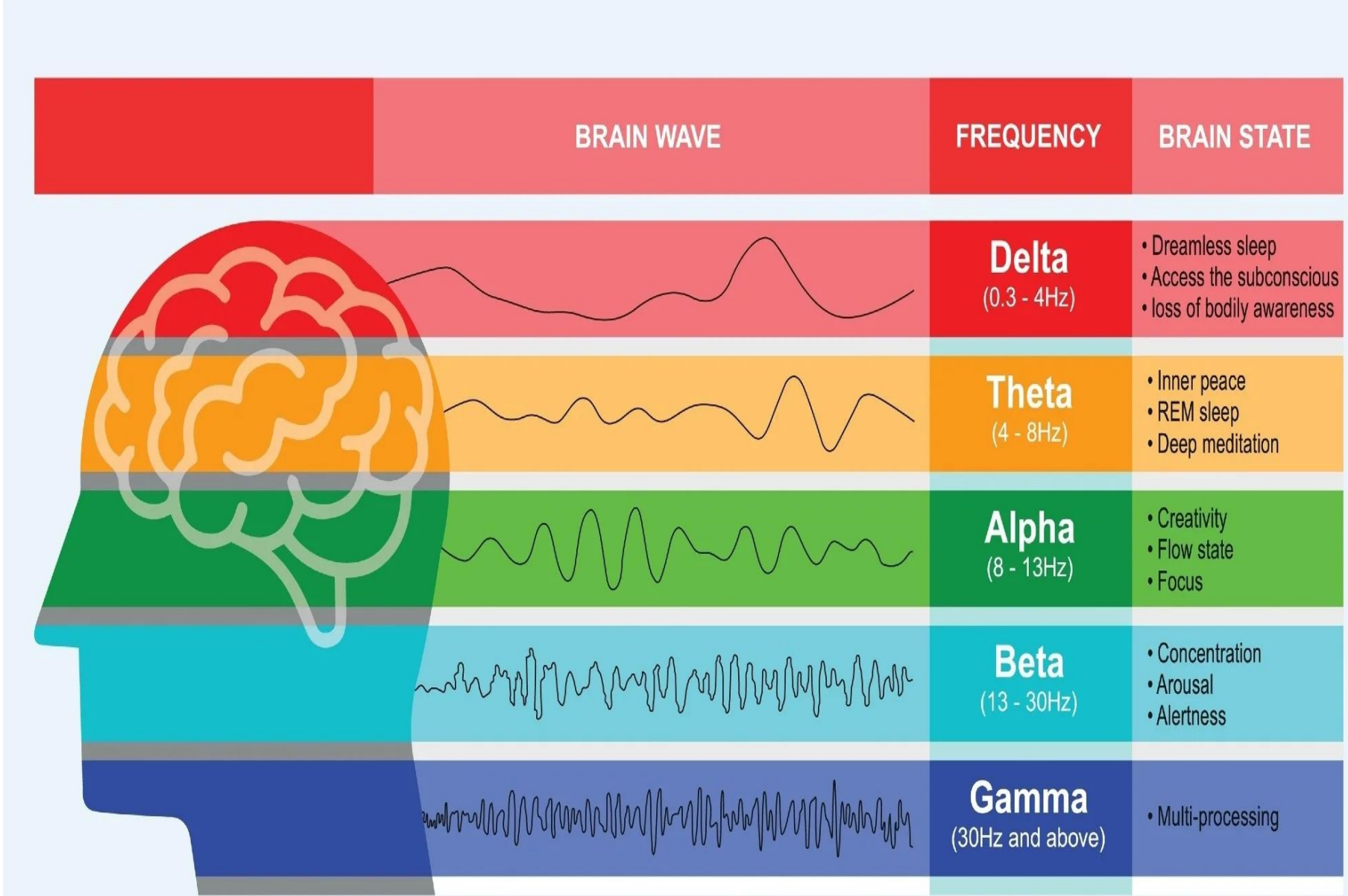
EEG Waves

Alpha wave

Beta wave

Theta wave

Delta waves



Main Types of Brain Waves

□ Delta Waves (0.5–4 Hz):

The slowest and highest-amplitude waves, “typically linked to deep sleep. They are normal in infants or during deep sleep in adults.

□ Theta Waves (4–7 Hz):

Slower frequency waves “often associated with daydreaming or meditation. They also appear during light sleep and drowsiness.

□ Alpha Waves (8–13 Hz):

Moderate frequency waves that are “common when you're awake but relaxed, especially with eyes closed. They originate mostly in the occipital region.

□ Beta Waves (13–30 Hz):

Higher frequency and lower amplitude waves present “when you're awake and thinking. They reflect active concentration and problem-solving.

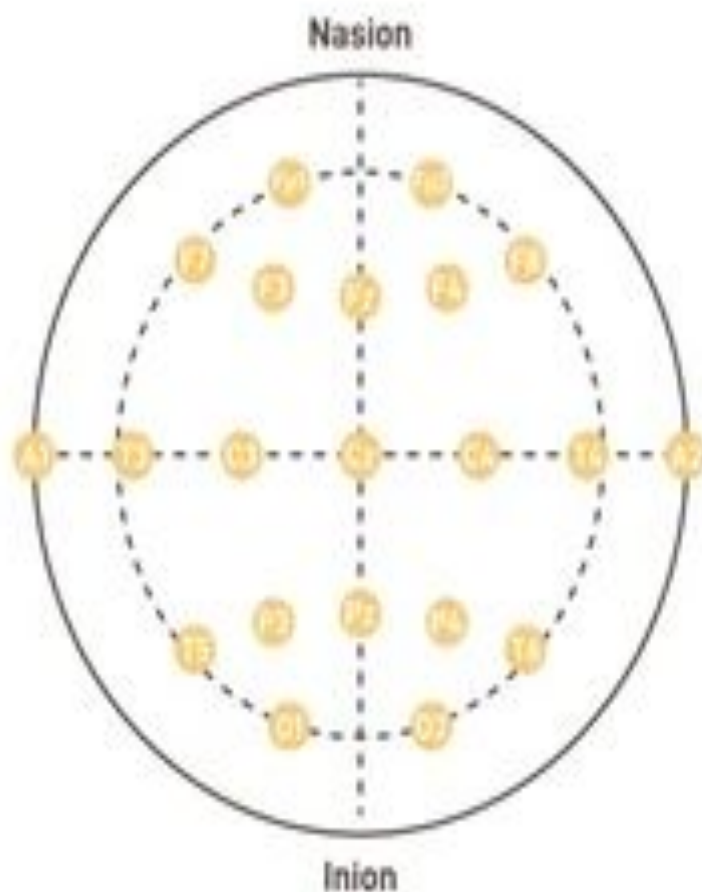
□ Gamma Waves (Greater than 30 Hz):

The fastest oscillations, which are “often present during deep focus.”

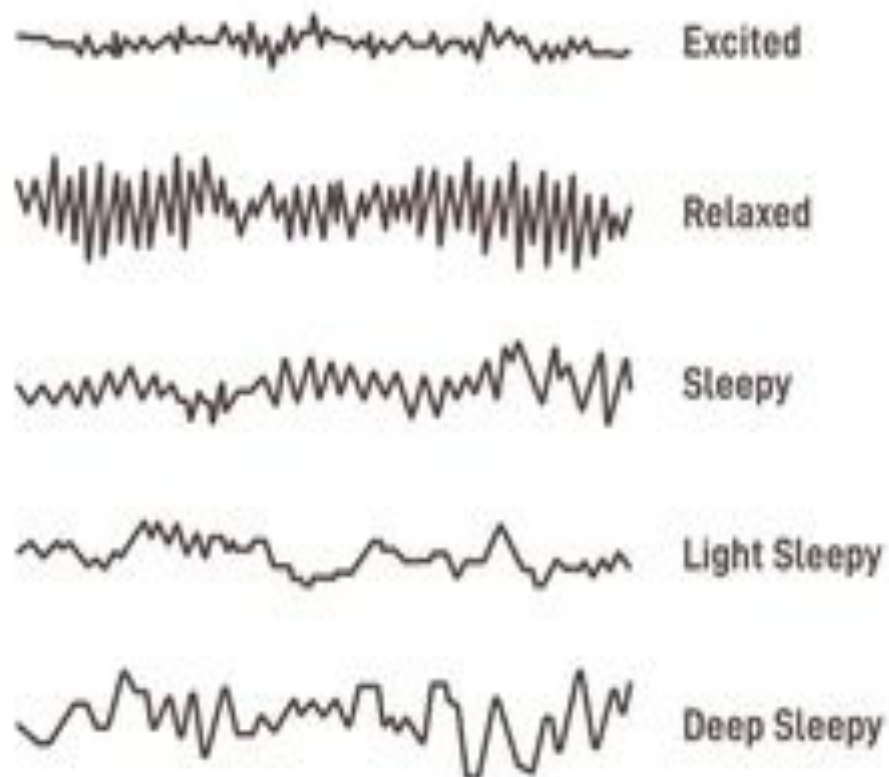
Different types of brain waves in normal EEG

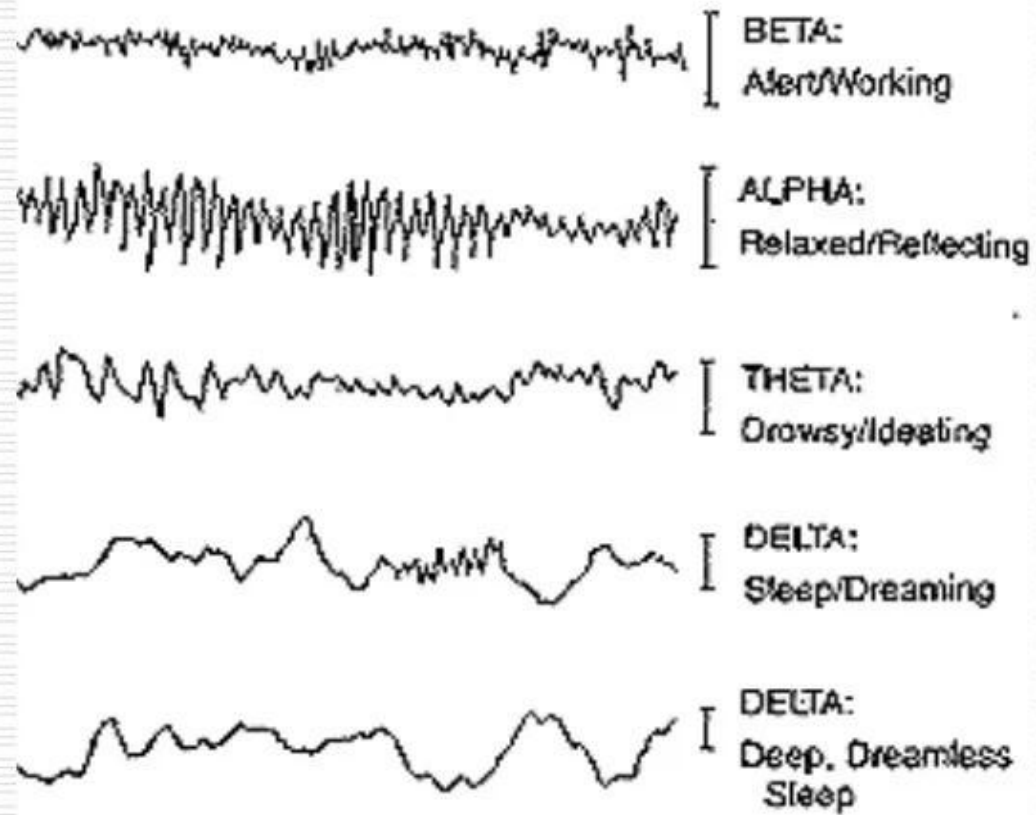
Rhythm	Frequency (Hz)	Amplitude (uV)	Recording & Location
Alpha(α)	8 - 13	50 - 100	Adults, rest, eyes closed. Occipital region
Beta(β)	14 - 30	20	Adult, mental activity Frontal region
Theta(θ)	4 - 7	Above 50	Children, drowsy adult, emotional distress Occipital
Delta(δ)	2 - 4	Above 50	Children in sleep

Electroencephalography

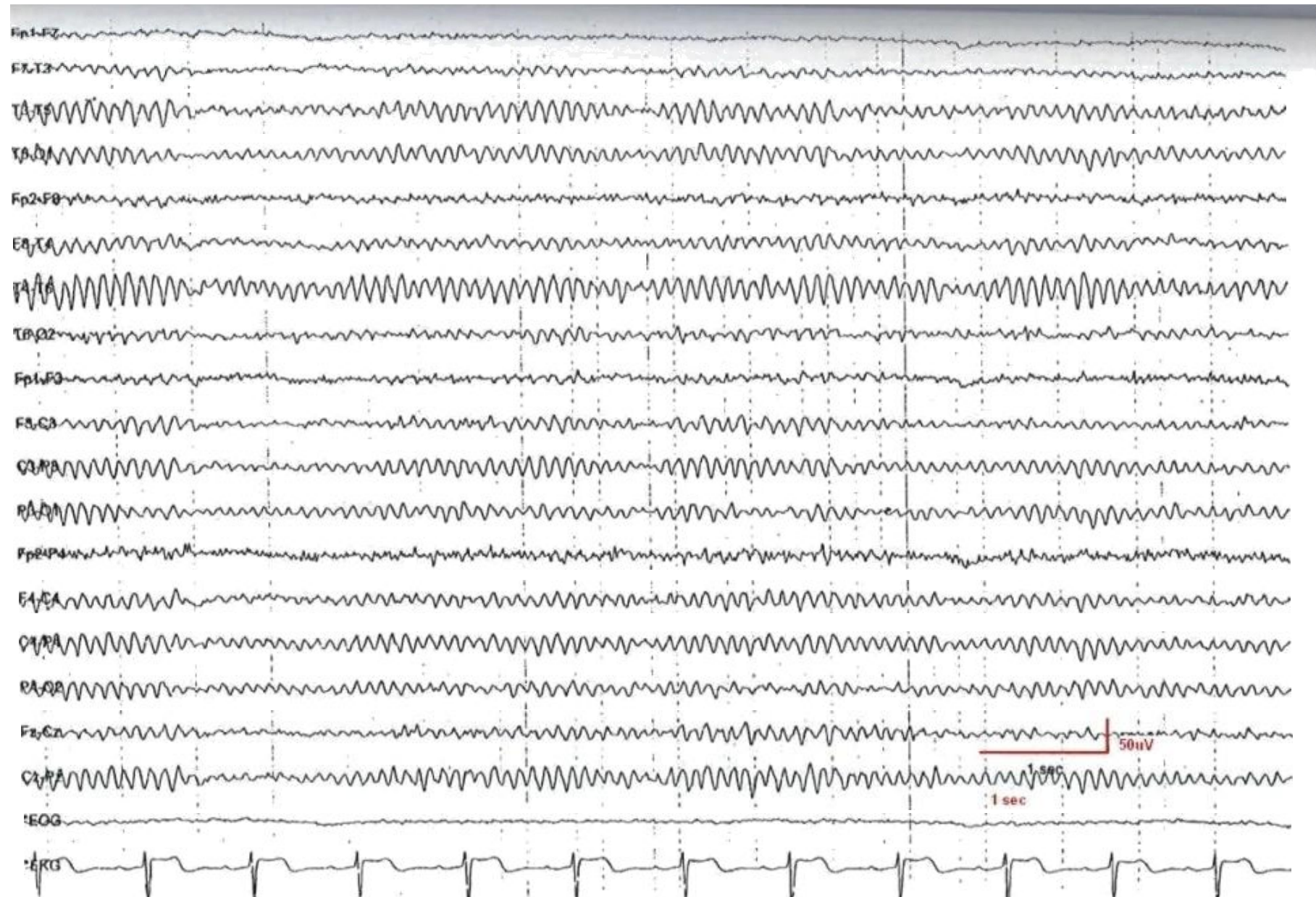


Brain Waves Chart





EEG Recording From Normal Adult Plate



Factor influencing EEG

Age

- + Infancy — theta, delta wave
- + Child — alpha formation.
- + Adult — all four waves.

/// Level of consciousness (sleep)

/// Hypocapnia(hyperventilation) slow & high amplitude waves.

Hypoglycemia

Hypothermia

Low glucocorticoids



5 low waves

Use of EEG

- Epilepsy
- Generalized (grandmal) seizures. Absence (petitmal) seizures.
- Localize brain tumors.
- Sleep disorders (Polysomnography)
 - Narcolepsy
 - Sleep apnea syndrome
 - Insomnia and
 - parasomnia
- Helpful in knowing the cortical activity, toxicity, hypoxia and encephalopathy
- Determination of brain death.
- Flat EEG(absence of electrical activity) on two records run 24 hrs apal.



EEG Test helps to detect-

- Seizure disorders like epilepsy
- Sleep disorders
- Dementia
- Head injury and concussion
- Encephalitis

Analysis

Electrical activity from the brain consist of primarily of rhythms.

They are named according to their frequencies (Hz) and amplitude in micro volt (gv).

Different rhythms at different ages and different conditions (level of consciousness)

The process of recording and examining electrical signals from the brain to diagnose medical conditions, understand neural activity, or power brain-computer interfaces.

- **Delta waves (up to 4 Hz):** Linked to deep sleep and regeneration.
- **Theta waves (4–8 Hz):** Linked to deep relaxation and emotional processing.
- **Alpha waves (8–13 Hz):** Linked to relaxed alertness and calm focus.
- **Beta waves (13–30 Hz):** Linked to active thinking and problem-solving.
- **Gamma waves (above 30 Hz):** Linked to high concentration and peak cognitive functioning.

Pathological EEG findings

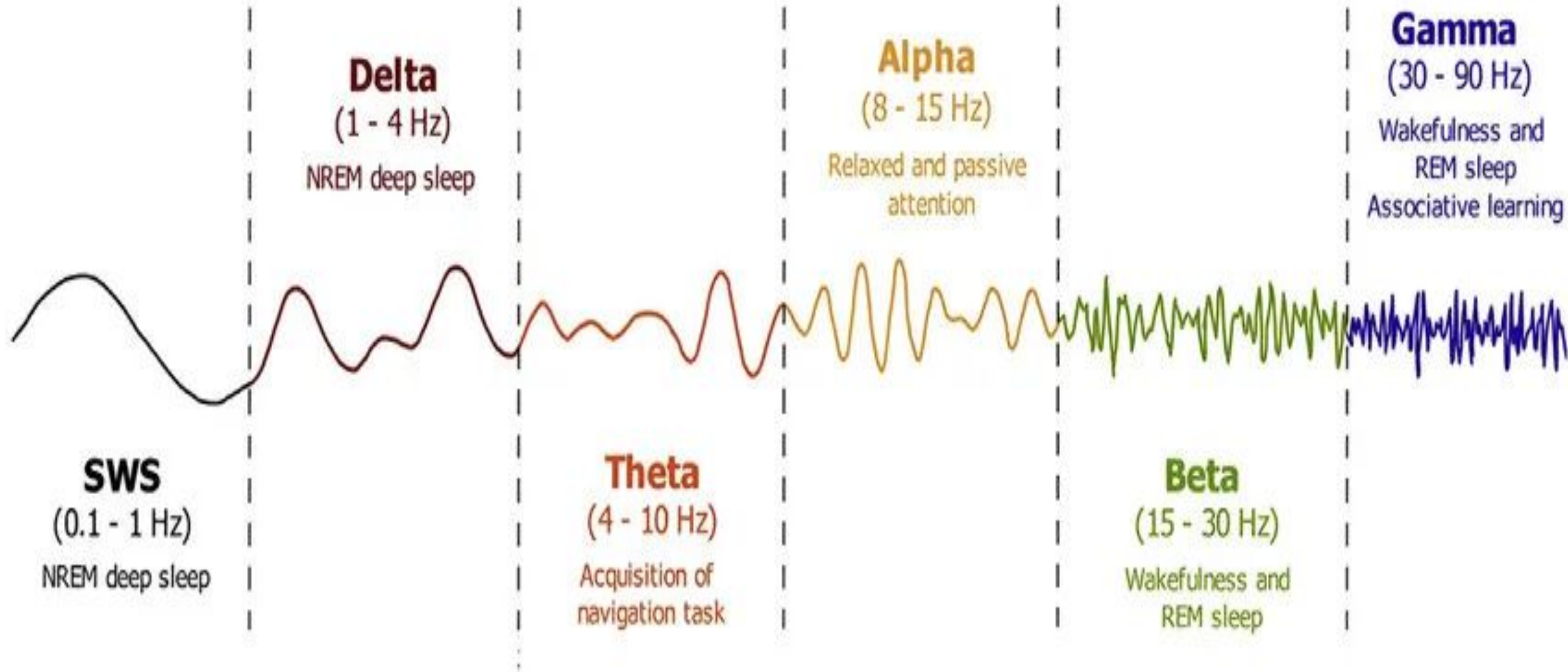
Increased delta power during NREM sleep in severe obstructive sleep apnea (correlates with apnea-hypopnea index, arousal index, oxygen desaturation)

Interictal epileptiform discharges (spikes, sharp waves) may indicate seizure disorder

Diffuse slowing suggests encephalopathy or structural abnormality

Reduced sleep spindle density is associated with cognitive impairment and Alzheimer disease

Brain waves



Requirements

EEG machine (8/16 channels).

Silver cup electrodes/metallic bridge electrodes.

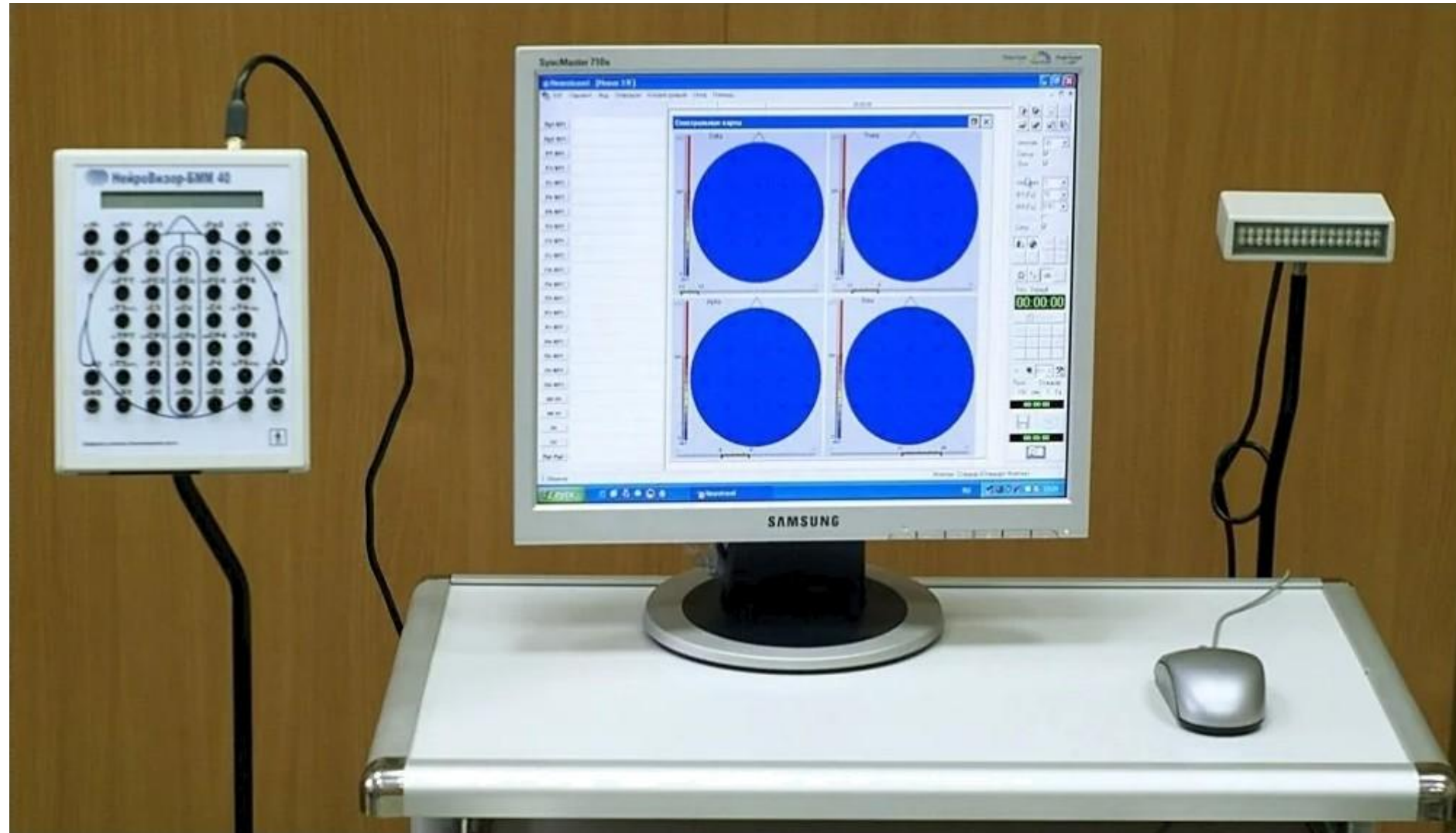
Electrode jelly.

Rubber cap.

Quiet dark comfortable room.

Skin pencil & measuring tape.

Computerized EEG



Electrode Positioning System



EEG Electrodes

Each electrode site is labeled with a letter and a number.

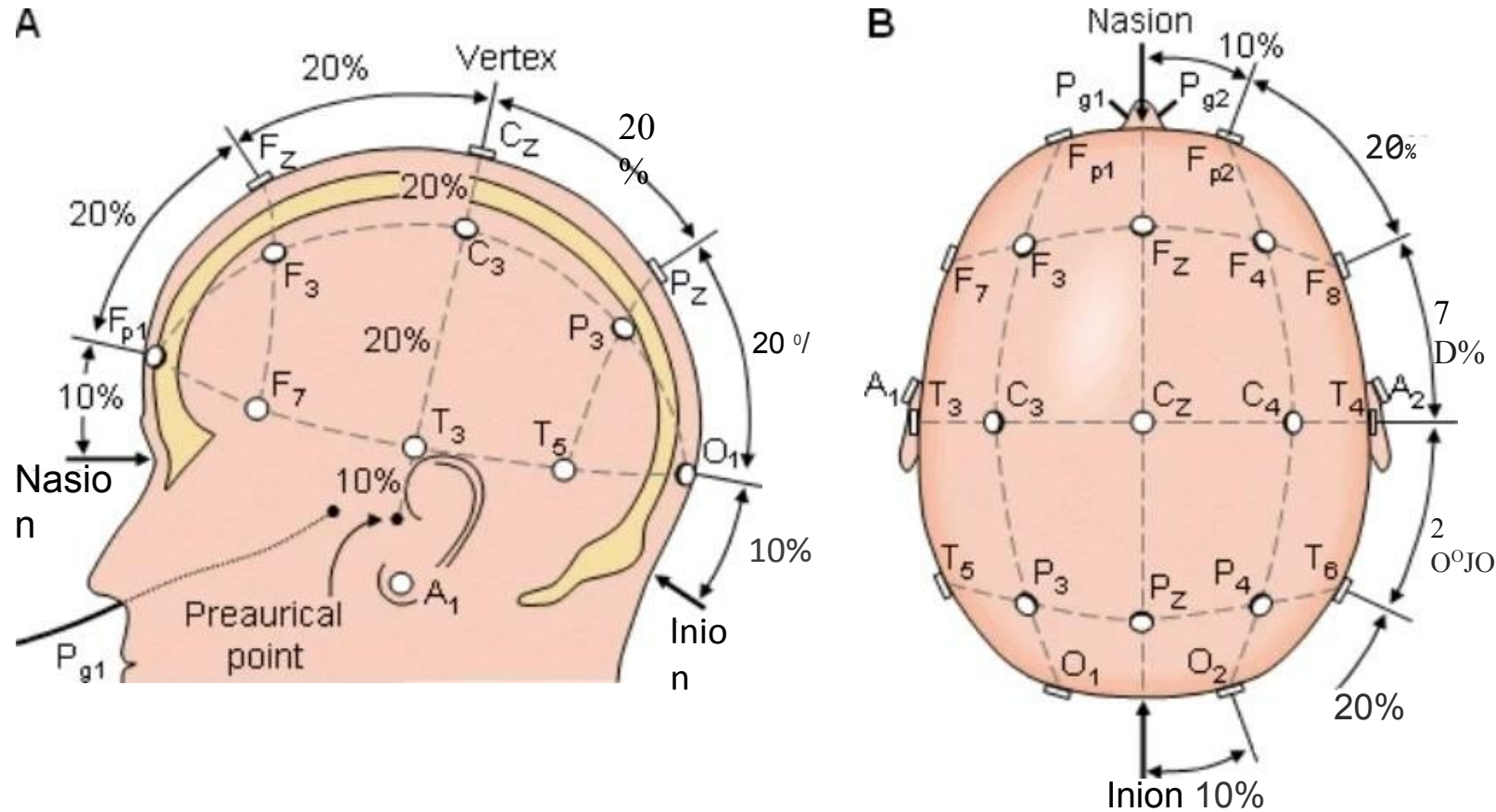
The letter refers to the area of brain underlying the electrode

e.g. F - Frontal lobe and T - Temporal lobe.

Even numbers denote the right side of the head

and Odd numbers the left side of the head.

20 % system of EEG electrode Placement



Procedure of EEG recording

- Press the calibration knob to check voltages & time constant.
- Always observe subject for any abnormal muscle activity.
- Ask the subject to open eyes for 10 sec. and ask him/her to close eyes. (do this procedure for several times in each montage)

Two types of recording

Bipolar-- both the electrodes are at active site

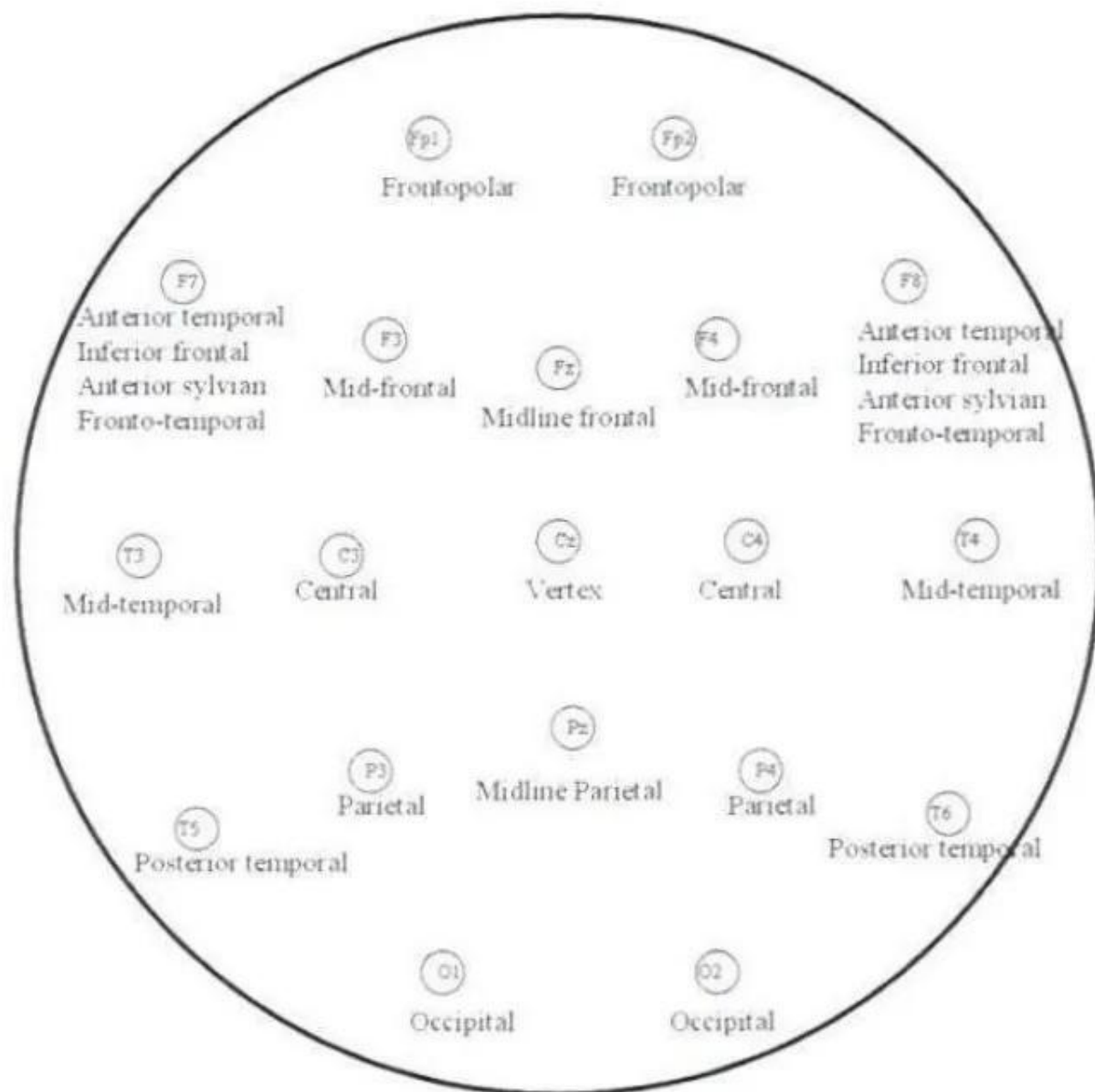
+ Bipolar montage are parasagittal montage.

Unipolar--one electrode is active and the other is in different kept at ear lobe.

Always watch for any abnormal muscle activity.

Ask the subject to open eyes for 10 sec. then ask them to close the eyes.

Full names for electrodes



EEG (Electroencephalogram)

Definition

- diagnostic test that records electrical activity of the brain using electrodes placed on scalp

Purpose / Indications

- Epilepsy/Seizure disorders
- Head injury
- Infections (e.g, encephalitis, meningitis)
- Sleep disorders
- Coma assessment
- Brain death confirmation

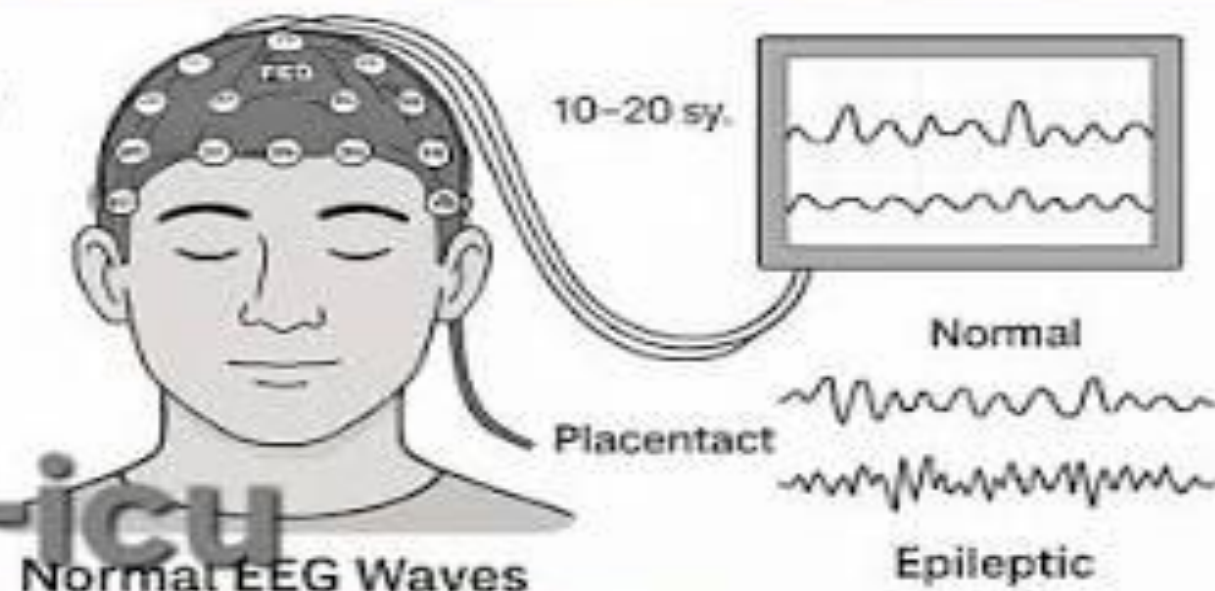
Procedure

- Electrodes frced to scalp with conductive
- Ask patient to relax, sometimes perform breathing exercises (hyperventilation) or be exposed to light (photostimulation)
- Brain waves recorded

Abnormal Findings

- Spikes / Sharp waves
- Focal slowing

Epilepsy
Encephathy
Flat EEG



Normal EEG Waves

Wave Type	Freq. (Hz)	Location
Alpha	8-13	Occipital
Beta	13-30	Frontal
Theta	4-7	Temporal
Delta	<4	Frontal

Abnormal Findings

- Spikes / Sharp waves: Epilepsy
- Diffuse slowing: Encephalopathy
- Focal slowing: Tumor/lesion

Preparation & Nursing Considerations

- Wash hair before (no oils, sprays)
- Avoid caffeine or sedative before
- Explain scientes)